

Abstract Algebra

: Lecture 10

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Today we talk about group actions.

✓ **Example 1.** Recall: Let $G = \text{GL}(V)$, this is a linear group. And we have a set V , also a vector space. → all invertible mat.s.

1. Each $g \in G$ is a bijection from V to V ; ✓.

2. $\forall v \in V$ and $1 \in G$ we have $v^1 = v$; ✓.

3. $\forall v \in V$ and $g, h \in G$ we have $v^{gh} = (v^g)^h$.

We called G acts on V .

$$\begin{matrix} K^m & \xrightarrow{A} & K^n & \xrightarrow{B} & K^l \\ x & \xrightarrow{\quad} & & & \end{matrix} \quad BAx = x^{AB} = (x^A)^B.$$

Definition 2. Let G be a group and Ω be a set. We say that G acts on Ω if

(1). each element of G is a bijection from Ω to Ω ;

(2). $\forall \omega \in \Omega, \omega^1 = \omega$;

(3). $\forall \omega \in \Omega$ and $g, h \in G$ we have $\omega^{gh} = (\omega^g)^h$.

We denote this action by $G \curvearrowright \Omega$.

G is a group composed of bij. from Ω to Ω .
i.e. $(hg)\omega = h(g\omega)$.

Example 3. $\text{Sym}(\Omega)$ and $\text{Alt}(\Omega)$ acts on Ω naturally. ✓.

$$\text{Sym}(\Omega) \curvearrowright \Omega, \text{Alt}(\Omega) \curvearrowright \Omega.$$

Example 4. Let G be a group, for each element $g \in G$, we define action $g : x \mapsto xg$ for all $x \in G$. We denote this map by $\hat{g}$, called right multiplication. This is a group action. We denote the group of this action by $\hat{G}$.

$$\hat{G} = \{ \hat{g} \mid \hat{g} : x \mapsto xg \text{ for all } x \in G, \text{ where } g \in G \}. \quad \rightarrow x^{\hat{g}} = g^{-1}xg.$$

Example 5. Let G be a group, for each element $g \in G$, we define action $g : x \mapsto g^{-1}xg$ for all $x \in G$. We denote this map by $\tilde{g}$, called conjugation. This is a group action. We denote the group of this action by $\tilde{G}$.

$$\tilde{G} = \{ \tilde{g} \mid \tilde{g} : x \mapsto g^{-1}xg \text{ for all } x \in G, \text{ where } g \in G \}.$$

Example 6. Let G be a group, for each element $g \in G$, we define action $g : x \mapsto g^{-1}x$ for all $x \in G$. We denote this map by $\check{g}$, called left multiplication. This is a group action. We denote the group of this action by $\check{G}$.

$$\check{G} = \{ \check{g} \mid \check{g} : x \mapsto g^{-1}x \text{ for all } x \in G, \text{ where } g \in G \}.$$

Exercise 7. If we write group action in another way, i.e. $(gh)\omega = g(h\omega)$, check example 4, 5, 6.

Proposition 8. 1. $\hat{g}, \tilde{g}, \check{g} \in \text{Sym}(G)$.

2. $\hat{G}, \tilde{G}, \check{G} \leq \text{Sym}(G)$.

$$(\hat{g}\hat{h})\omega = \omega \cdot (h \cdot g) = (\omega \cdot h) \cdot g = \hat{g}(\hat{h}\omega).$$

$$(\tilde{g}\tilde{h})\omega = \tilde{g}^{-1}h^{-1}\omega hg = g^{-1}(\tilde{h}\omega)g = \tilde{g}(\tilde{h}\omega).$$

$$(\check{g}\check{h})\omega = \check{g}^{-1}h^{-1}\omega = \check{g}^{-1}(\check{h}\omega) = \check{g}(\check{h}\omega).$$

$$xg = g^{-1}xg \text{ for some } g.$$

$$x=1 \Rightarrow g=g^{-1}g=1$$

$$\Rightarrow \bar{g} = \bar{1}.$$

$$3. \forall g, h \in G, \hat{g}\hat{h} = \hat{h}\hat{g}.$$

$$4. \forall g \in G, \hat{g}\hat{g} = \bar{g};$$

$$5. \hat{G} \cap \check{G} = Z(\hat{G}) = Z(\check{G}), \hat{G} \cap \check{G} = \check{G} \cap \hat{G} = \{1\};$$

$$6. \langle \hat{G}, \check{G} \rangle = \hat{G}\check{G} = \hat{G} \circ \check{G} = \hat{G} : \check{G} = \check{G} : \hat{G};$$

$$\hat{G} \cap \check{G} = \{ \bar{g} \mid \bar{g} : x \mapsto xg, xg = g^{-1}x \text{ for some } g \}.$$

$$x\bar{g}\bar{g} = (xg)\bar{g} = (g^{-1}x)\bar{g} = g^{-1}xg \in \check{G}. \Rightarrow g^{-1}xg = x \Rightarrow gx = xg$$

$$\text{for all } x \in G. g \in Z(G). g^{-1} \in Z(G) \Rightarrow \hat{G} \cap \check{G} = Z(\hat{G}) = Z(\check{G}).$$

$$Z(\hat{G}) = \{ \bar{g} \mid \bar{g} : x \mapsto xg = gx \} = Z(\check{G})$$

$$C \in (G \times H), C \cong A \in Z(G) \times \{1\} \cong Z(G).$$

Definition 9. For two groups G and H , assume there exist $C \leq Z(G)$ and $C \leq Z(H)$. s.t. $C \neq \{1\}$.

Let $Z_1 \leq Z(G)$ and $Z_2 \leq Z(H)$. s.t. $Z_1 \simeq Z_2 \simeq C$. Let ϕ be an isomorphism from Z_1 to Z_2 . Let

$X = (G \times H) / \langle (x, x^\phi) \mid x \in Z_1 \rangle \simeq (G \times H) / C$. This group X is called a **central product of G and H** ,

denoted by $G \circ H$.

Definition 10. Let $H, K < G$ s.t. $H \triangleleft G$ and $H \cap K = \{1\}$. Then $\langle H, K \rangle = HK = H \rtimes K = H : K$

called a **semi-direct product of H and K**

G acts on Ω also can be said as an action of G on Ω or group action of G on Ω .

$$\text{Orb}(\omega).$$

Definition 11. Let G act on Ω . Then G partitions Ω into orbits, where an orbit is $\Delta = \omega^G = \{\omega^g \mid g \in G\}$,

where $\omega \in \Omega$. So $\Omega = \bigsqcup_{\omega \in \Omega} \omega^G$.

$$\omega_1 \sim \omega_2 \Leftrightarrow \exists g \in G \text{ st } g(\omega_1) = \omega_2. (\omega_1 \bar{g} = \omega_2)$$

Example 12. $\tilde{G}$ acts on G naturally, i.e. $x^g = g^{-1}xg$. $G = x_0^G \sqcup x_1^G \sqcup \dots \sqcup x_t^G$, where $x_0 = e$,

$x_i^G = \{g^{-1}x_i g \mid g \in G\}$ called a **conjugacy class of x_i** , denoted by $C(x_i)$. $C(x_i) = \{g^{-1}x_i g \mid g \in G\}$ is a

orbit of the action of $\tilde{G}$ on G . $\checkmark$.

Definition 13. For G acting on Ω , $G_\omega = \{g \in G \mid \omega^g = \omega\}$ called the **stabilizer of ω in G** . It's easy to check that G_ω is a subgroup of G . $\checkmark$.

$$|Orb(\omega)| = |G : Stab(\omega)| \Leftrightarrow |G| = |Orb(\omega)| |Stab(\omega)|$$

Theorem 14. (Orbit-Stabilizer Theorem) For G acting on Ω , For $\omega \in \Omega$, $|G| = |\omega^G| \cdot |G_\omega|$.

证明. Let $\Delta = \omega^G = \{\delta_1, \delta_2, \dots, \delta_m\}$ write $\delta = \delta_1$. Let g_i s.t. $\delta^{g_i} = \delta_i$ for $1 \leq i \leq m$. $\checkmark$.

Claim: For any element $x \in G$, $\delta^x = \delta_i \Leftrightarrow x \in G_\delta g_i$. Which is due to $\delta^x = \delta_i = \delta^{g_i} \Leftrightarrow \delta^{xg_i^{-1}} = \delta \Leftrightarrow xg_i^{-1} \in G_\delta \Leftrightarrow x \in G_\delta g_i$.

Recall $G = G_\delta \sqcup G_\delta g_2 \sqcup \dots \sqcup G_\delta g_m$, we have $|G| = |G_\delta| + |G_\delta g_2| + \dots + |G_\delta g_m|$, i.e. $|G| = |\Delta| |G_\delta|$. $\square$

Observe: consider conjugate action of G on itself, $|x^G| = 1$ iff $x \in Z(G)$. So by Orbit-Stabilizer

Theorem, $|G| = |Z(G)| + |C(g_1)| + \dots + |C(g_r)|$ and $|g_i^G| \mid |G|$ due to $|G| = |C(g_i)| \cdot |C_G(g_i)|$.

Theorem 15. (Sylow's 1st Theorem) Let G be a finite group, $|G| = p^e m$ s.t. p is a prime and

$(p, m) = 1$, a subgroup H of G s.t. $|H| = p^e$ exists. And H is called a **Sylow p -subgroup** of G , denoted

by $H \in Syl_p(G)$.

Recall: If G is abelian, then let $H = \{g \in G \mid |g| = p^e\}$. Then H is a subgroup of G and $|H| = p^e$.

In other words H is a Sylow p -subgroup of G .

$$\textcircled{1} \text{ induction on } |G|. |G| = p^n m.$$

证明. Write $|G| = |Z(G)| + |C(g_1)| + \dots + |C(g_r)|$. If $p \mid |Z(G)|$ then $Z(G)$ has a Sylow p -subgroup N and $N \triangleleft G$. Then $\bar{G} = G/N$ has order $|\bar{G}| < |G|$. If $|N| = p^e$ then N is a Sylow p -subgroup of G . If $|N| < p^e$ then $|\bar{G}| < p^e m$. By induction, $\bar{G}$ has a Sylow p -subgroup $\bar{N}$, the preimage of $\bar{N}$ is a Sylow p -subgroup of G .

$\text{Orb}(g_i)$, but is also a conjugacy class.
 $\text{Stab}(g_i)$.

Now suppose $p \nmid |Z(G)|$. Then $p \nmid |C(g_i)|$ for some i . By Orbit-Stabilizer Theorem, $|G| = |C(g_i)| \cdot |C_G(g_i)|$. Then $p^e \mid |C_G(g_i)|$ and by induction $C_G(g_i)$ has a Sylow p -subgroup N , which is also a Sylow p -subgroup of G . $\square$

Theorem 16. (Cauchy) If $p \mid |G|$ then G has a subgroup of order p . $\checkmark$

Lemma 17. If G is a p -group i.e. $|G| = p^n$ then G has a non-trivial center. (Not Sylow's law yet!).

9. This exercise outlines a proof of Cauchy's Theorem due to James McKay (Another proof of Cauchy's group theorem, Amer. Math. Monthly, 66(1959), p. 119). Let G be a finite group and let p be a prime dividing $|G|$. Let $\mathcal{S}$ denote the set of p -tuples of elements of G the product of whose coordinates is 1:

$\mathcal{S} = \{(x_1, x_2, \dots, x_p) \mid x_i \in G \text{ and } x_1 x_2 \cdots x_p = 1\}$.
ordered, $x_i \dots x_p x_1 \dots x_{i-1}$

- (a) Show that $\mathcal{S}$ has $|G|^{p-1}$ elements, hence has order divisible by p .
 $x_{i-1} x_i (x_i \dots x_p)^{-1} = x_1 \dots x_{i-1}$

Define the relation $\sim$ on $\mathcal{S}$ by letting $\alpha \sim \beta$ if β is a cyclic permutation of α .

- (b) Show that a cyclic permutation of an element of $\mathcal{S}$ is again an element of $\mathcal{S}$. $\checkmark$
 (c) Prove that $\sim$ is an equivalence relation on $\mathcal{S}$. $\checkmark$
 (d) Prove that an equivalence class contains a single element if and only if it is of the form $(x, x, \dots, x)$ with $x^p = 1$. $\checkmark$
 (e) Prove that every equivalence class has order 1 or p (this uses the fact that p is a prime). Deduce that $|G|^{p-1} = k + pd$, where k is the number of classes of size 1 and d is the number of classes of size p . $\checkmark$
 (f) Since $\{(1, 1, \dots, 1)\}$ is an equivalence class of size 1, conclude from (e) that there must be a nonidentity element x in G with $x^p = 1$, i.e., G contains an element of order p . [Show $p \nmid k$ and so $k > 1$.] Otherwise, $p \mid \text{LHS}$, $\text{RHS} \equiv 0 \pmod{p}$. $\checkmark$